

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: COMPUTER NETWORKS COURSE CODE:
CSA07**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

*I SANDHIYA S(192521224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
SANDHIYA S

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router.

The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANSWERS

1. Given IP block: 192.168.100.0/24

A /24 network contains 256 IP addresses (192.168.100.0 – 192.168.100.255). Using VLSM (Variable Length Subnet Masking), allocate the largest subnet first and proceed in descending order.

1. School of Computing – /25

- Subnet Mask: 255.255.255.128
- Block Size: 128 addresses
- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127
- Valid Host Range: 192.168.100.1 – 192.168.100.126
- Usable Hosts: 126

2. School of Electronics – /26

- Subnet Mask: 255.255.255.192
- Block Size: 64 addresses
- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191
- Valid Host Range: 192.168.100.129 – 192.168.100.190
- Usable Hosts: 62

3. School of Mechanical Engineering – /27

- Subnet Mask: 255.255.255.224
- Block Size: 32 addresses

- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range: 192.168.100.193 – 192.168.100.222
- Usable Hosts: 30

4. Remaining Unused IP Address Range

After allocating the above subnets:

- Allocated range:
 - 192.168.100.0 – 192.168.100.223
- Remaining range:
 - 192.168.100.224 – 192.168.100.255

This unused portion forms a /27 subnet:

- Network Address: 192.168.100.224
- Broadcast Address: 192.168.100.255
- Valid Host Range: 192.168.100.225 – 192.168.100.254
- Usable Hosts: 30

Department	CIDR	Subnet Mask	Network Address	Host Range	Broadcast Address	Usable Hosts
School of Computing	/25	255.255.255.128	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126

School of Electronics	/26	255.255.255.192	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.1	62
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School of Mechanical Engineering	/27	255.255.255.224	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.2	30
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Remaining Unused Block	/27	255.255.255.224	192.168.100.224	192.168.100.225 – 192.168.100.254	192.168.100.2	30
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Total usable hosts allocated:

$$126 + 62 + 30 = 218 \text{ hosts}$$

Remaining usable hosts available:

30 hosts

Conclusion: The IP block 192.168.100.0/24 has been successfully subnetted into /25, /26, and /27 networks for the three departments, leaving one unused /27 subnet available for future expansion.

2.

Given Network Block: 10.10.0.0/16

The allocated network 10.10.0.0/16 contains 65,536 IP addresses ranging from 10.10.0.0 to 10.10.255.255.

Using VLSM, the subnets are allocated in descending order of size:

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Subnet Calculations

1. Software Development (/24)

- Subnet Mask: 255.255.255.0
- Network Address: 10.10.0.0
- Broadcast Address: 10.10.0.255
- Host Range: 10.10.0.1 – 10.10.0.254
- Usable Hosts: 254

2. Quality Assurance (/25)

- Subnet Mask: 255.255.255.128
- Network Address: 10.10.1.0
- Broadcast Address: 10.10.1.127
- Host Range: 10.10.1.1 – 10.10.1.126
- Usable Hosts: 126

3. Human Resources (/26)

- Subnet Mask: 255.255.255.192
- Network Address: 10.10.1.128
- Broadcast Address: 10.10.1.191
- Host Range: 10.10.1.129 – 10.10.1.190
- Usable Hosts: 62

4. Executive Management (/27)

- Subnet Mask: 255.255.255.224
- Network Address: 10.10.1.192
- Broadcast Address: 10.10.1.223
- Host Range: 10.10.1.193 – 10.10.1.222
- Usable Hosts: 30

Remaining Unused IP Address Range

Allocated addresses occupy:

- 10.10.0.0 – 10.10.1.223

Therefore, the remaining unused range is:

- 10.10.1.224 – 10.10.255.255

Department	CIDR Prefix	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
Software Development	/24	255.255.255.0	10.10.0.0	10.10.0.1	10.10.0.254	10.10.0.255	254
Quality Assurance	/25	255.255.255.128	10.10.1.0	10.10.1.1	10.10.1.126	10.10.1.127	126

Human Resources	/26	255.255.255.192	10.10.1.1 28	10.10.1.1 29	10.10.1.190	10.10.1.191	62
Executive Management	/27	255.255.255.224	10.10.1.1 92	10.10.1.1 93	10.10.1.222	10.10.1.223	30
Remaining Unused Range	N/A	N/A	10.10.1.2 24	10.10.1.2 25	10.10.255.2 54	10.10.255.2 55	65,05 6

Final Answer

- Total usable hosts allocated: $254 + 126 + 62 + 30 = 472$ hosts
- Unused IP range: 10.10.1.224 – 10.10.255.255
- The company still has a large portion of the 10.10.0.0/16 address space available for future departmental expansion.

3. Given IPv6 Network

2001:0db8:abcd:0000:0000:0000:0000/64

1. Convert the Given IPv6 Address into Compressed Notation

IPv6 compression rules:

- Leading zeros in each block can be omitted.
- Consecutive blocks of zeros can be replaced with :: (only once).

Given Address:

2001:0db8:abcd:0000:0000:0000:0000

Compressed Form:

2001:db8:abcd::

2. Expand the Compressed Address Back to Full Representation

Compressed Address:

2001:db8:abcd::

Expanded Form:

2001:0db8:abcd:0000:0000:0000:0000

3. Identify the Network Prefix Using /64 Prefix Length

IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000/64

A /64 prefix means:

- First 64 bits = Network Prefix
- Last 64 bits = Interface Identifier (Host Portion)

Network Prefix:

2001:0db8:abcd:0000::/64

or in compressed form:

2001:db8:abcd:0::/64

4. Calculate the Total Number of Host Addresses Supported Within the Subnet

For a /64 network:

- Total IPv6 bits = 128

- Network bits = 64
- Host bits = 64

Number of Host Addresses:

$$2^{64} - 2 = 18,446,744,073,709,551,616$$

Total host addresses supported = 18,446,744,073,709,551,616

5. Total Number of Possible Host Addresses Available Within This Network

Since the network is a /64 subnet, the available host addresses are:

$$2^{64} - 2 = 18,446,744,073,709,551,616$$

Total possible host addresses available = 18,446,744,073,709,551,616

Task	Result
Given IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000/64
Compressed Notation	2001:db8:abcd::
Expanded Notation	2001:0db8:abcd:0000:0000:0000:0000
Network Prefix (/64)	2001:0db8:abcd:0000::/64

Host Portion Length	64 bits
Formula for Host Addresses	2^{64}
Total Host Addresses Supported	18,446,744,073,709,551,616
Total Possible Host Addresses Available	18,446,744,073,709,551,616

Conclusion

The IPv6 address 2001:0db8:abcd:0000:0000:0000:0000/64 compresses to 2001:db8:abcd::/64. With a /64 prefix, the first 64 bits represent the network portion and the remaining 64 bits are used for host addressing, allowing 18,446,744,073,709,551,616 unique host addresses within the subnet.

4. Given Routing Table

Network Address	Prefix Length
192.168.1.0	/24
192.168.2.0	/24
192.168.3.0	/24

Destination IP Address

192.168.2.45

1. Determine the Destination Subnet

The routing table contains the subnet:

192.168.2.0/24

A /24 subnet uses the subnet mask 255.255.255.0, meaning:

- Network Address: 192.168.2.0
- Host Range: 192.168.2.1 – 192.168.2.254
- Broadcast Address: 192.168.2.255

Since 192.168.2.45 falls within this range, the packet belongs to:

Destination Subnet: 192.168.2.0/24

2. Matching Routing Table Entry

The router compares the destination IP address against its routing table entries.

Destination IP:

192.168.2.45

Matching Route:

192.168.2.0/24

This is the best match according to the Longest Prefix Match rule.

3. Next-Hop Network/Interface Used

Since the destination matches 192.168.2.0/24, the packet is forwarded through the interface (or next hop) associated with that route.

Next-Hop Network/Interface: Interface connected to network 192.168.2.0/24

4. Default Route if No Match Exists

If the destination IP does not match any routing table entry, the router uses the default route.

Default Route:

0.0.0.0/0

This route forwards packets to the configured default gateway (next-hop router).

Example:

- Destination IP: 192.168.10.50
- No matching route found
- Router forwards packet using 0.0.0.0/0

Item		Result
Destination IP Address	192.168.2.45	
Destination Subnet	192.168.2.0/24	
Matching Routing Table Entry	192.168.2.0/24	
Subnet Mask	255.255.255.0	

Valid Host Range	192.168.2.1 – 192.168.2.254
Next-Hop Network/Interface	Interface connected to 192.168.2.0/24
Default Route (if no match exists)	0.0.0.0/0
Action Taken by Router	Forward packet to the interface associated with 192.168.2.0/24

Conclusion

The destination IP address 192.168.2.45 belongs to the subnet 192.168.2.0/24. The router matches the routing table entry 192.168.2.0/24 and forwards the packet through the interface connected to that network. If no matching route exists, the router uses the default route (0.0.0.0/0) to forward the packet to the default gateway.

5. Given Information

Department	Network
Administration LAN	192.168.10.0/26
Finance LAN	192.168.10.64/26

Router Interfaces

Interface	IP Address
Administration Gateway	192.168.10.1
Finance Gateway	192.168.10.65

1. Determine the Network Address and Broadcast Address

Administration LAN (192.168.10.0/26)

- Subnet Mask: 255.255.255.192
- Block Size: 64 addresses
- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63

Finance LAN (192.168.10.64/26)

- Subnet Mask: 255.255.255.192
- Block Size: 64 addresses
- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127

2. Valid Host IP Address Range

Administration LAN

- First Host: 192.168.10.1
- Last Host: 192.168.10.62
- Valid Host Range: 192.168.10.1 – 192.168.10.62

- Usable Hosts: 62

Finance LAN

- First Host: 192.168.10.65
- Last Host: 192.168.10.126
- Valid Host Range: 192.168.10.65 – 192.168.10.126
- Usable Hosts: 62

3. Suggested IP Addresses for Three Computers in Each LAN

Administration LAN

Computer	IP Address
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Admin-PC1	192.168.10.2
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Admin-PC2	192.168.10.3
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Admin-PC3	192.168.10.4
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Finance LAN

Computer	IP Address
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Finance-PC1	192.168.10.66
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Finance-PC2 192.168.10.67

Finance-PC3 192.168.10.68

4. Default Gateway for Each LAN

Hosts must use the router interface connected to their LAN as the default gateway.

LAN	Default Gateway
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Administration LAN	192.168.10.1
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Finance LAN	192.168.10.65
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Department	Network Address	Subnet Mask	First Host	Last Host	Broadcast Address	Usable Hosts	Default Gateway
Administration LAN	192.168.1.0	255.255.255.192 (/26)	192.168.1.1	192.168.1.62	192.168.1.63	62	192.168.1.1
Finance LAN	192.168.1.64	255.255.255.192 (/26)	192.168.1.65	192.168.1.126	192.168.1.127	62	192.168.1.65

Suggested Computer IP Addresses

Department	Computer	IP Address
Administration	Admin-PC1	192.168.10.2
Administration	Admin-PC2	192.168.10.3
Administration	Admin-PC3	192.168.10.4
Finance	Finance-PC1	192.168.10.66

Finance	Finance-PC2	192.168.10.67
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Finance	Finance-PC3	192.168.10.68
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Conclusion

The Administration LAN (192.168.10.0/26) uses 192.168.10.1 as its default gateway and supports hosts from 192.168.10.1 to 192.168.10.62. The Finance LAN (192.168.10.64/26) uses 192.168.10.65 as its default gateway and supports hosts from 192.168.10.65 to 192.168.10.126. The suggested computer addresses are valid and allow communication both within each LAN and between departments through the Layer-3 router.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

